Coding structure: Synthesis, Categories, Subcategories			Description	Examples of data excerpts
Synthesis 1 Familiarization	AI onboarding (C1)	Building AI knowledge (SC1)	Systematic explanation of AI concepts, principles, application potential, and usage strategies by teacher educators.	For the two cohorts in this study, the module began with an overview of AI, providing definitions of important concepts such as AI, GenAI, and LLMs. (No.10, p. 142) During the first five weeks of the practicum, the participants were taught the fundamental principles of using AI in language instruction, including practical applications such as utilizing ChatGPT, Pi, Sider and Copilot for lesson planning and tutoring. (No. 12, p. 178) Workshop 1 consisted of two key components. One was a macro-level introduction to the concept of ChatGPT, its applications in education, and prompt-giving principles for interacting with it (Spasi´c & Jankovi´c, 2023). The other covered fundamental concepts and key principles of writing feedback, such as its purpose and key considerations (Hattie & Timperley, 2007). (No. 29, p. 6)
		Modelling AI practices (SC2)	Indirect AI usage experiences through modelling by experts, teacher educators, or peers using AI.	The teacher educators understood that if they want their students to use GenAI tools responsibly and critically, they must model how to use them in their classrooms and expose their students to different uses. For example, Steven mentioned, 'We need to demonstrate to students how to use AI. We might model some ideas of how to use AI to generate ideas or generate feedback that [students] can use to complete your assignment or something. I think students might need to be explicitly shown how to use it properly.' (No.20, p.6) Additionally, the classroom demonstrations by teachers during workshops were instrumental in helping PSTs understand the overall process of using ChatGPT for writing feedback. PST01 reflected, "Although I knew the steps to communicate with ChatGPT, without the classroom demonstrations, I would have been struggling for a long time." Similarly, PST04 supplemented, "Before the class, I felt I couldn't use ChatGPT effectively. After the teacher's demonstration in workshops, I realized how

				to communicate with ChatGPT properly, which gave me many ideas and boosted my confidence in using ChatGPT." (No. 29, p. 11) Meanwhile, PSTs reported spending substantial time observing, discussing, and comparing their prompts and initial feedback with those of their peers and teacher demonstrations. The significant impact of vicarious experience on self-efficacy may be attributed to the novelty of providing feedback on student writing for PSTs (Bandura, 1977). Given the lack of an objective standard for the quality of feedback generated by PSTs during practice activities, individuals had to evaluate their abilities by observing and comparing themselves with others (Bautista & Boone, 2015). (No. 29, p. 12-13) We then presented a five-minute hands-on demonstration video illustrating the process of using ChatGPT to provide writing feedback in alignment with these principles. The video highlights three key steps: making an initial assessment of the student's essay, interacting with ChatGPT, and making strategic revisions. After that, the participants engaged in group activities to practice these steps, complete worksheets, and share their experiences. They could repeatedly watch the video during the group activities for reference and imitation. (No. 29, p.6)
	AI-supported scenarios (C2)	Experiencing affordances of AI tools (SC3)	Presentation of AI tools and their functions to establish tool-function-context mappings.	Teacher 10's Week 8 reflection demonstrated this integration: My systematic learning approach has helped me effectively use AI tools. I now use ChatGPT for initial differentiated task creation, Perplexity for cultural content, Magic School for visual grammar explanations, and Twee, Magic School, Dift.me, and ChatGPT for practice activities. I now understand each tool's strengths and combine them to meet my diverse student needs. (No. 14, p. 21234) By Week 4, her entries showed emerging competence in both TPACK and SRL strategies, demonstrating the role of microlearning in structuring her learning process: teaching applications:

				I have set specific goals for each AI tool and use them systematically to generate level-appropriate materials. Today, I used Alayna.us for lesson planning, Perplexity for researching authentic materials, and Magic School for creating listening comprehension tasks. (No. 14, p. 21235) Teacher 18's interview revealed: If ChatGPT's writing feedback were not detailed enough, I would try Magic School's approach. If Perplexity's cultural content requires adaptation, I refine it using Twee's level-specific features. This helped me develop expertise in matching tools to specific teaching needs. (No. 14, 21237) The second half of this module focused on the pedagogical implications and applications of AI. TCs examined various AI tools and applications for creating educational content, such as lesson plans, unit plans, teaching resources, and assessment tools. For example, TCs explored Eduaide and Magic School to develop lesson plans, unit plans, and teaching resources; Differ to facilitate differentiated instruction; Quizziz for creating assessments; and Microsoft coaches as an example of intelligent tutoring and adaptive assessment tools. (No. 10, p. 142) For example, they used ChatGPT to refine teacher talk and generate evaluative feedback, and adopted Copilot to develop interactive activities such as image-based storytelling and visual vocabulary journals, as illustrated in the following refection: I made classroom activities that can make English learning more interesting to students using ChatGPT and Copilot. For example, ChatGPT suggested a visual vocabulary journal activity, where students can work with Copilot to make images with text prompts. I believe this activity can enhance not only their English skills but also their creativity. (Refection paper, Participant 11). (No. 18, p. 4-5)
		AI case-based instruction (SC4)	Review and analysis of AI-integrated instructional cases	Between the lecture and case-based discussion, in which teachers are prompted to develop instructional solutions, we recommend introducing additional model cases in which the problems and solutions are presented [60]. We further recommend

			demonstrating AI use in teaching contexts.	presenting well-structured problems, such as rule-using problems, in themodel cases to help teachers see how new theories and tools can be applied in contexts. The additional model of cases would help teachers bridge the gap between theory and practice and produce AI-integrated instructional solutions in case-based discussions with ill-structured problems. (No. 8, p. 11) Before introducing this prompt, participants were provided with an example to illustrate the importance of context in shaping responses. The example the grading scale in Austria ranges from 1 to ___ was displayed via WebSocket on participants' screens while the lecturer explained the context. This was designed to demonstrate how providing additional contextual information influences the specificity and accuracy of responses. Participants were not required to respond to this example; instead, it served to prepare them for understanding the role of context. (No. 4, p. 20)
	AI progressive exploration (C3)	Scaffolded exploration (SC5)	Guided AI task completion under structured support (e.g., worksheets, checklists, explicit task instructions).	Additionally, we provided a checklist (see Appendix D) summarizing the principles and strategies covered in the session as a scaffold to support their learning and help them internalize the knowledge during the practice. (No. 29, p. 6) During the intervention, participants were provided with model prompts and allowed to fine-tune their interactions until they achieved the outcomes they desired. Applying this skill to language teaching, they can craft prompts based on the specific task (e.g. creating an activity about a grammar point), age, proficiency level, and topic, leading to more refined and valuable outputs (Maloy & Gattupalli, 2024). (No. 19, p. 9) Initially, participants focused on understanding fundamental GenAI capabilities, and fourteen reported that the bite-sized format helped them overcome AI anxiety. Teacher 7 noted in Week 2: Breaking down GenAI learning into small chunks makes it less daunting. Today, I focused on designing communicative activities, learning to prompt ChatGPT, and practicing group tasks with Twee. This targeted approach helped me see how AI can enhance my lesson planning. (No. 14, p. 21236)

		Open exploration (SC6)	Opportunities for open exploration of AI tools based on individual interests, encouraging experimentation and discovery.	Moreover, allowing student teachers to interact with AI for their interests, and exposing student teachers to various AI are useful activities for the Exploration phase. Data analysis indicates that these activities allow student teachers to experiment with AIs. As STUDENT TEACHER XX reflected, "I learned how a variety of AIs can be utilized in the spheres of academic and research writing. The use of ChatGPT, perplexity, Elicit, and Connected papers to search and consolidate relevant data was amazing" Based on this reflection, it is evident that these activities were helpful for student teachers to determine the strengths and weaknesses of each AI appropriately. (No.24, p. 23946) The hands-on practice offered PSTs the chance to apply ChatGPT in specific contexts, allowing them to engage with the technology directly and thereby enhancing their confidence in proficiency with ChatGPT. For example, PST01 remarked, "When I applied what I learned in practice, I became more familiar with how to interact with ChatGPT and engage in dialogue to achieve the desired outcomes." The cumulative effect generated by repeated practice also proved this point. PST06 added, "I believe I can do better because we have had many opportunities to practice… I feel that in the process of providing ChatGPT-assisted feedback, I have been able to improve, gradually doing better and becoming more efficient." (No. 29, p. 10) Second, schools should allocate dedicated time for teachers to engage with GAI in a hands-on capacity, facilitating a deeper exploration of its capabilities and ensuring that teachers can see the practical value of the technology. (No. 3, p. 22701)
Synthesis 2 Reflective practice	Reflective practice on AI (C4)	Role negotiation (SC7)	Reflection on teacher-AI role boundaries and professional identity in AI-enhanced teaching contexts.	… (during the AI-enhanced training), I really saw the potential of AI to amplify our shared knowledge. But here's the thing-it struck me that no matter how advanced the tech gets, it shouldn't overshadow the personal connections we've built in our classrooms. …. It's this blend of new skills from the training and my own teaching style that really shapes my approach. I'm all for integrating AI, but in a way that

				complements, not replaces, the personal interaction that makes teaching so rewarding.…' (Leo, interview). (No. 17, p.7) The participants learned to be curators of technology tools outputs, not just consumers. Another participant (S3) wrote that he struggled at first because the automatically created problems "lacked the specific context I wanted for my class, so I had to tweak the wording and setup." He concluded that "Technology tools can give a great starting point, but the teacher needs to shape it into a meaningful task." These reflections reinforce that the educational context and teacher input are crucial – a finding aligned with prior studies that warn about unguided technology tools usage [1], [14]. (No. 22, p. 128968) Teachers' dual roles of prompting and critical reviewing. After the intervention, the responses from PSTs suggested the potential of Gen-AI to enhance the quality of feedback provided to students. However, they also maintained that this technology did not diminish the crucial role played by human teachers in the educational process. Instead, PSTs positioned teachers as critical reviewers of AI-generated content, emphasizing the importance of human checking and revision in improving feedback quality. For example, PST03 highlighted, "After ChatGPT has assessed the writing sample, it must be reviewed by humans again, based on the Gaokao rubric criteria." PST02 provided a more detailed perspective, noting, "ChatGPT sometimes follows English-speaking logic, which can lead to issues in sentence expression, and I will choose to correct them myself." (No. 29, p. 10)
		Output evaluation (SC8)	Critical reflection on AI-generated instructional materials.	It was stressed that AI is a support tool, not a replacement for the teacher's own thinking. Throughout the course, pre-service teachers were instructed to use AI-generated outputs critically – for example, checking for accuracy and alignment with curricular goals – rather than copy-pasting content. (No. 22, p. 128970). Such critical reflection is particularly crucial when using generative AI – it prompted the participants to scrutinize the AI's suggestions for biases or inaccuracies and to articulate their own rationale in lesson planning. By reflecting on the AI's role, pre-

				service teachers cultivated a habit of questioning and contextualizing AI outputs, thereby reinforcing their critical thinking and lifelong learning skills. (No. 22, p. 128969) They then generate images based on prompts like “a person,” “a woman,” “a house,” “a street,” and “a plate of food,” incorporating variations with cultural or geographical descriptors. This step aims to uncover how the AI interprets these prompts and whether it introduces cultural biases. After generating the images, educators collaboratively analyze how AI systems interpret human input, fostering a deeper understanding of inherent biases and enabling them to critique AI-generated speculations in light of their own lived experiences. (No. 7, p. 299). All educator reflections revealed ongoing concerns about the ethical implications of biased data and the reinforcement of stereotypes. These activities appeared to have opened up space for critical reflection by prompting educators to think more deeply about the complexities of AI integration in education. (No. 2, p. 304) Cognitive changes also impact how PSTs employ ChatGPT-generated content to offer their feedback. During the training workshop, PSTs were requested to revise ChatGPT-generated initial feedback when providing feedback on student essays (see Fig. 2). It is evident that by adding to or modifying ChatGPT’s responses, PSTs transformed vague feedback into more precise and actionable comments. For example, while ChatGPT suggested, “In terms of creativity, try to propose some novel ideas or unique solutions,” One PST provided more specific feedback: “In terms of creativity, try to propose some novel ideas or personal reflections, such as how I would approach the situation, to bring the reader closer.” (No. 29, p. 10)
		Documented reflection (SC9)	Documentation and reflection on AI use through structured artefacts (e.g., reflective journals, portfolios, concept maps, blog entries).	The incorporation of structured reflection activities proved highly valuable in this study. This study found that the reflection journals were the medium through which pre-service teachers solidified their understanding of the role of AI tools in teaching. This reinforces the broader principle that reflection is a critical component of teacher education, especially when new technologies are involved. By reflecting, participants

				moved beyond the excitement of using a new gadget and critically analyzed its pedagogical value. (No. 22, p. 128970) Phase 1: Weekly Reflective Journals. Throughout the 13-week course, participants maintained structured reflective journals to document their experiences with GenAI tools, learning strategies and teaching applications. Guided prompts encouraged detailed narratives, focusing on the following: (No. 14, p. 21231-21232) • Learning experiences with the weekly modules • Application of GenAI tools in ELT contexts • Challenges encountered and solutions developed • SRL strategies employed • Plans for future implementation Participants submitted their journals weekly, allowing us to continually track how their use of GenAI tools for language teaching developed. Submitting the journals every two weeks allowed us to monitor their growth (e.g. Moon, 2006) using an iterative analytic approach. My refection log helps me identify the best tool combinations for different teaching objectives. (No. 14, p. 21236)
		Reflective discussion (SC10)	Group discussions foster critical reflection through multiple perspectives.	A pivotal element in our collaborative process was the role of each other as critical friend. Acting as critical friends, we provided each other with constructive feedback, challenged each other's assumptions, and offered diverse perspectives that enriched our understanding of GenAI technologies. (No. 25, p. 14) Regarding instruction, the teacher educators were acutely aware that they can play critical roles in modelling the use of GenAI tools and engaging their students in reflective discussions about the implications of these tools on their learning and future professional practices. (No. 20, p. 8)

				The activity also fosters critical reflection as participants assess AI's impact on teaching and learning. By actively engaging in the discussion and mapping process, educators collaboratively shape their understanding of AI integration in education and build knowledge together. (No. 7, p. 299)
		AI-mediated reflection (SC11)	Reflective conversation with AI systems for automated support and immediate feedback.	In the absence of a mentor, novice teachers had the opportunity to enhance their pedagogical practice and knowledge with the assistance of ChatGPT, as NT4 said, 'Since experienced teachers, mentors, or supervisors were not always present, it was important to make some decisions individually'. In such cases, novice EFL teachers could professionally experience a socio-constructivist learning approach, have automated support and immediate feedback, and generate knowledge and sharpen their skills through reflection with ChatGPT, as the first researcher observed during discussion groups: 'Novice teachers discerned that they used to explain and talk a lot during their teaching since ChatGPT highlighted the role of students' inclusion and engagement'. (No. 2, p. 8) PSTs resorted to GenAI tools such as ChatGPT and POE for pedagogical advice, grammar check, and content creation. They relied heavily on GenAI tools as experienced teachers and experts, providing feedback for them and as creative content generators creating learning and teaching materials. For instance, Flora found it convenient to use GenAI tools because her professor suggested PSTs to use ChatGPT or POE to enhance their writing. (No. 28, p. 6)
Synthesis 3 Collaborative design	Collaboration (C5)	Peer support (SC12)	Building AI competencies and skills through peer interactions within learning.	Finally, some participants (n = 6) identified group work as providing peer scaffolding in developing the technological skills needed for GenAI-enhanced lesson planning. In fact, personal innovativeness—referring to individual differences in GenAI literacy in this study—was identified as an important factor in obtaining quality outputs from GenAI. Some participants experienced challenges in composing effective prompts for GenAI, which made it difficult to achieve their desired results. To compensate for this personal lack of technological skills, they chose to utilize group collaboration as scaffolding to generate quality outputs from GenAI tools:

				While working on lesson plans with GenAI, I sometimes struggled to get the exact results I wanted because I believe I wasn't as familiar with GenAI as my classmates were. Then, I chose to collaborate with other classmates to form a group. I could depend on them to refine the prompts and improve the outputs. (No. 18, p. 5) We learned that student teachers can find a better pedagogical approach if they work in groups created based on their specialization. (No. 24, p. 23946) In addition, the training program is suggested to emphasize vicarious learning through group discussions and collaborative activities, as our findings suggest that high-quality vicarious learning experiences can be particularly impactful in shaping PSTs' self-efficacy and skills within a short period. This balance can foster both operational proficiency and reflective thinking, enabling PSTs to integrate Gen-AI effectively into their feedback practices. (No.29, p.13)
		AI resource & strategy sharing (SC13)	Sharing of AI experiences, instructional materials, prompting strategies, and design outcomes.	To improve my skills using GenAI tools, I often found online forums and social media groups where other teachers shared tips and strategies. For example, I found a tutorial on YouTube that helped me refine my prompts, and then I could create better lesson plans. (No. 18, p. 6) While educators share their individual perspectives about how they envision the future educational setting, they also engage in a collective inquiry process by interacting with each other's alternative futures. These AI-generated collages catalyze discussions on educators' perspectives regarding the role of AIEd, focusing on topics such as how AI might transform teaching and learning, its impact on the role of teachers, and broader societal implications. (No.7, p. 298-299)
		Community support (SC14)	Establishment of sustained learning communities for ongoing mutual support, resource sharing, and collective growth.	Additionally, building a global community for educators to share insights and strategies can further explore the dual focus on AI technology and the preservation of teacher-student rapport (Ghafouri, 2024). This collaboration and exchange of knowledge across regions aim to advance educational technology collectively. It highlights the collaborative essence of the HIAIC model and its ability to foster a

				more inclusive and human-centred approach to technology adoption in education worldwide. (No. 17, p. 11) Community support was provided for lesson reflection, offering suggestions and feedback to enhance teaching effectiveness. This support allowed the teachers to learn how to make adjustments to address the challenges encountered when they implemented their AI-based lesson plans……Community support plays a vital role in helping teachers tackle challenges and facilitating efficient adjustments in instructional strategies during lesson enactment. Teachers' beliefs in classroom management, student engagement, and instructional strategies significantly improve as students actively use technology and demonstrate progress in English speaking accuracy. (No. 26, p. 3) Receiving community support (e.g., feedback and suggestions for the reflections on teaching experiences and the modification of lesson plans) provided by a senior engineer and college teachers with expertise in technology-assisted language learning during practical teaching (No. 26, p.3)
	AI-supported design (C6)	Co-creation with AI (SC15)	Co-creation of new instructional content with AI tools, emphasizing teacher-AI collaboration.	First, novice and experienced EFL teachers interacted with ChatGPT individually before their class to plan their lessons, receive guidance, prepare authentic materials, and know how to teach the specific content (see Appendix 1 for sample teacher interactions with ChatGPT and Appendix 2 for a sample lesson plan co-designed with ChatGPT). (No. 2, p. 5) - The participants noted that AI significantly aided them in developing efficient lesson plans. "AI has been quite effective in helping me develop an interactive and interpersonal lesson plan." (S15) AI is capable of providing teachers with several lesson plans at the same times. This can help teachers to have plan A and Plan B for their teaching easily. (No. 12, p. 180)

		AI-informed pedagogical adaptation (SC16)	Adaptation and contextualization of AI-generated or existing materials to align with instructional goals, learner levels, cultural contexts, and curriculum standards.	My skills to use ChatGPT have been improved, but I believe the ELT knowledge is also so important to get refined results, because it can make the outputs well-suited for my lessons. (No. 18, p. 5) Each pre-service teacher was tasked with designing a comprehensive lesson plan for a secondary math topic (of their choice or assigned) while integrating MATH41. They formulated learning objectives and then used MATH41 to generate practice problems, examples, or visual aids aligned with those objectives. During this period, they iteratively refined their lesson plans. (No.22 p. 128964) This evolution is exemplified by Ms. Lee, who adeptly adapted song lyrics to meet her language learners' comprehension levels. Similarly, Ms. Fang's methodical content adaptation process involved utilizing ChatGPT to refine educational song lyrics, transforming them to align with lesson objectives and making them more accessible to EFL learners. (No. 27, p.7) I asked ChatGPT 3.5 to create a reading comprehension activity for the book Dumplings for Lily (Iwai, 2021). After my first two prompts, I found the ChatGPT-generated reading comprehension prompts were too simplistic for the grade levels I specified. They missed the cultural and linguistic considerations. I would have to prompt the chat box with these requests so it would include these elements. (No. 25, p.10)
		Iterative prompting (SC17)	Iterative prompting and the refinement of prompts through multi-turn dialogue to develop effective AI interaction skills.	As part of the activity instructions, TCs asked the applications to revise the AI-generated lesson plans multiple times. TCs were encouraged to chat with the tools and not take the first generated output. TCs did the same for creating a formative assessment tool, a summative assessment tool, and a rubric for assessment….. TCs said: AI can be an extremely useful tool, but you need to know how to use it properly. It also takes more than one prompt to get the outcome we want. We should utilize several prompts and not settle for the first one. (No. 10, p. 142-144)

				The participants reflected that their iterative interactions with GenAI tools through repeated prompting and editing work enhanced their GenAI literacy; that is, their perceptions and effective use of GenAI affordances developed through their collaboration with GenAI tools. This ongoing engagement with GenAI not only developed their technical skills but also boosted their self-efficacy. Their reflections further showed that their pedagogical knowledge, as well as their improved technological skills, contributed to refining GenAI outputs for use in lesson planning: As I worked more with ChatGPT, I found that my skills get better and better over time. It's like the more I refined prompts to ChatGPT outputs, the more likely I'm to create useful lesson content, which in turn made me feel more confident in my tech skills. (No. 18, p. 5) Increased engagement with ChatGPT is also crucial for enhancing feedback quality. This iterative process involves refining prompts through ongoing interaction, leading to more effective and tailored feedback. Common interactions include follow-up questions, further inquiries based on ChatGPT-generated content, and requesting ChatGPT to modify its generated content. For example, instead of using a general prompt like "What is your comment on this essay?" (Fig. 1), PST01 learned to request specific feedback on areas such as content, expression, and developmental level after the intervention. During the interviews, PST01 explained, "If I continuously pose follow-up questions to ChatGPT, it will, of course, go a step further in searching for answers, which will certainly enhance the quality of the feedback provided." (No. 29, p. 10)
Synthesis 4 A cycle of authentic teaching	AI-integrated teaching cycle (C7)	AI-assisted microteaching (SC18)	Microteaching activities involving AI-assisted instruction conducted in supportive and low-risk environments (e.g., workshops) to practice and refine teaching approaches.	Participants conducted a 20-minute segment of their designed lesson in a microteaching session. In these sessions, each pre-service teacher taught their lesson excerpt to their peers (who role-played as students) They utilized the materials created with MATH41 during instruction. (No. 22, 128964) The influence of micro-teaching sessions was evident from student teachers' reflections. As STUDENT TEACHER XXI reflected, one experiencing moment was

				an assignment where we went off as a group, because of not understanding it clearly. It was a reflecting moment whereby any teacher who wants to integrate technology into his/her teaching, has to ask himself/herself several questions before integration to enhance the teaching and learning practices. Therefore, this reflection shows how valuable the micro-teaching exercise challenged student teachers in a way that they needed to make informed judgments about the choice of their AI. (No. 24, p. 23947-23948) Workshops were conducted for five months (five sessions, 90 minutes each) to focus on lesson planning. These workshops included micro-teaching activities to help the teachers enhance their abilities to plan lessons integrated with the Questions app, including its application in English speaking instruction and the design of instructional activities. (No. 26, p. 3)
		AI-integrated classroom teaching (SC19)	Implementation of AI-assisted instruction in authentic classroom settings, involving sustained teaching practice with real students.	In lesson enactment, the teachers implemented lesson plans integrated with the Questions app in their weekly EFL instructions for three months (eight sessions, 40 minutes each). (No. 26, p. 3) We also learned the need to provide opportunities for teachers to experiment their lesson plans in the actual classroom contexts to gain more insights for dealing with diverse contextual needs. (No. 24, p. 23947) Weeks 5-7: Implementation (Experimentation Phase) The teachers implemented the CALL lessons they designed in their classrooms. This phase allowed them to experiment with AI integration and investigate the outcomes in an authentic setting. (No. 27, p. 5)

		Emotional enrichment of AI content (SC20)	Enhancing AI-generated content and feedback with emotional tone and human sensitivity to reduce mechanical or impersonal qualities.	Sam: I think, I think its (Inq-ITS) biggest problem, you know, it is not necessarily… it is not necessarily the instructional aspect of it. It is that… that sense of indifference, if you tell the kids you are wrong without telling them why, their pride is gone…. It is like there is one answer here. I think Rex (the intelligent tutor in Inq-ITS) increased their anxiety. The kids got frustrated that they were answering the question and kept getting told that they were wrong. They could not handle that. (No. 8, p. 7) TPST08 noted the importance of modification, "After ChatGPT responds, we still need to make some modifications and additions to its feedback … As teachers, we should add some subjective emotions when we give feedback to students, and we should not make our feedback look hard and cold." PST also embedded culturally resonant encouragement (e.g., "The essay includes numerous examples from literary classics as support, demonstrating the author's profound literary knowledge and extensive reading experience"). With the intervention and their practices, they can integrate ChatGPT's feedback input more thoughtfully, enhancing the human touch and warmth of their feedback, which ChatGPT alone could not generate. (No. 29, p. 10) Hua's experience with integrating virtual reality (VR) into teaching illustrates the struggle to maintain human connections amidst interactive technology. After discovering VR's potential during AI-enhanced training, Hua was enthusiastic but soon recognized its impact on reducing face-to-face interactions. This realization led Hua to consider how to balance technological innovation with personal engagement. This addresses the humanity-technology tension, ensuring that technology enhances rather than dilutes education quality. (No. 17, p. 9)
	AI-integrated teaching improvement (C8)	AI-integrated lesson refinement (SC21)	Systematic refinement of AI-integration in instruction.	Afterwards, groups of TCs critiqued the generated output by answering the following prompts: • Evaluate the generated products: Provide a short evaluation of the pros and cons. • What is missing in your lesson plan? What can be improved? • What are some barriers to learning in this lesson?

				• What else should we consider? • Reflect on the process of using AI tools to generate lesson plans and educational content. (No.10) After teaching, each participant wrote a detailed reflection journal about their lesson planning and teaching experience. Prompts in the reflection asked them to evaluate the effectiveness of their lesson design, how the AI tool and MATH41 contributed to or altered their approach, challenges faced, and what they would improve next time. Participants were encouraged to connect theory to practice, commenting on how using MATH41 influenced student engagement or understanding in their microteaching. (No. 22, p. 128964) The teachers evaluated the efficacy of their AI-integrated lessons by analyzing student progress and their observation notes. They shared insights, critiqued the effectiveness of their own and each other's methods, and refined their CALL lessons. (No. 27, p. 5)
		Responsible AI use (SC22)	Critical reflection on the appropriateness, ethical implications, and limitations of AI use in teaching.	The teachers' reflections on AI's ability to offer interactive feedback tailored to individual proficiency levels, while cautioning against an overdependence on AI, resonates with the call for a balanced and ethical integration of AI in educational settings (Prasad et al., 2022). Together, these insights advocate for a nuanced, critical approach to the adoption of AI in language education, ensuring that technological advancements serve to complement rather than supplant the pedagogical goals and human elements that lie at the heart of effective teaching and learning. (No. 27, p. 12) Equally important, professional development for teachers needs to include how they and their students can approach their use of AI tools critically, being aware of potential bias and inaccurate information (Kumar, 2023). While tools for generating

				lesson plans, for example, may be timesavers, they must also be used judiciously. (No. 9, p. 84) The ethics component was under discovery throughout all critical co-discovery activities and course modules. Educators were engaged in a collective inquiry process for various ethical concerns associated with AI in educational settings, such as potential dilemmas in AI integration, privacy issues, transparency, accountability in AI decision-making processes, and the risk of perpetuating inequalities or silencing minoritized voices. (No. 7, p. 304) ……Critical co-discovery activities offer educators a way to reflect on their autonomy in integrating AI and making pedagogical decisions in educational settings. (p. 306)
Synthesis 5 Affective dimension	Affective dimension (C9)	Emotional reactions (SC23)	Emotions experienced by teachers during AI use, including anxiety, worry, enjoyment, satisfaction, and other emotions.	Besides, reflections and interactions with ChatGPT created the space for novice teachers where they became more motivated, creative, and satisfied in developing their teaching knowledge and practice and emotionally felt better. (No.2, p.8) The findings revealed that educators in this study felt significant pressure to integrate AI into their teaching practices, leading to confusion and uncertainty, particularly since AI remains a relatively new concept for many. This sense of obligation suggests that teachers often perceive AI integration as a mandatory requirement (Lee et al., 2024). By fostering an environment that prioritizes exploration over enforcement, the critical co-discovery approach can help alleviate the anxiety associated with AI adoption and empower educators to make informed, confident decisions about when and how to incorporate AI into their classrooms. (No. 7, p. 308) Yet here's the thing. On one hand, as highlighted in one of the training sessions, I'm a big believer in the power of personal connections in education and the unique teaching style each teacher brings to the table. On the other hand, as we explored AI applications, there's nagging worry that we might start relying too much on tech, maybe even diluting those personal touches and individual teaching styles we

				cherish. The training forced me to confront this tension, … it's like I'm walking a tightrope ….' (No. 17, p. 7)
		Attitudes & beliefs (SC24)	Teachers' attitudes, perceptions, trust, and value beliefs toward AI tools.	Second, several TCs indicated that they left the module with more acceptance to explore the technology and better attitudes to embrace it (critically). For example: I think that the main takeaway from this session, along with discussions with other students, is that AI is a brave new world but can actually work as something we should embrace rather than shy away from. This can help save time on lesson creation time by providing a great base, but not a final product. (No. 10, p. 144) Similarly, PST07 noted, "Before the training, I viewed ChatGPT as a tool limited to basic tasks like checking punctuation, word choice, or grammar. Its potential to assist me in providing writing feedback seemed limited. However, after the training, I realized that with appropriate prompts, it can generate reliable and useful suggestions." This shift highlights a transformation from initial skepticism to appreciation, as PSTs now view ChatGPT as a valuable, supportive resource rather than a threat. (No. 29, p. 7) In efforts to support and study teachers' trust in AI-EdTech, it is paramount that the PDPs create conditions that are conducive to generating trust. (No. 21, p. 928) Another critical theme identified in our findings was the prevalence of false information regarding AI tools and their capabilities. Several participants reported feeling hesitant to adopt AI technologies due to misconceptions about their effectiveness or concerns about reliability. For instance, some preservice teachers believed that AI could replace traditional teaching methods entirely, leading to resistance in integrating these tools into their pedagogical practices. This highlights a significant gap in knowledge that teacher education programs must address. (No. 12, p. 184)
		Self-efficacy (SC25)	Teachers' confidence and self-assessment of their	As I worked more with ChatGPT, I found that my skills get better and better over time. It's like the more I refined prompts to ChatGPT outputs, the more

			competencies to use AI tools in teaching.	likely I'm to create useful lesson content, which in turn made me feel more confident in my tech skills. (No. 18, p. 5) In other words, under targeted intervention, PSTs' self-efficacy in using ChatGPT and the quality of writing feedback they provide form a mutually reinforcing positive cycle. As PSTs become more adept at leveraging ChatGPT, their growing confidence (related to self-efficacy) leads to more effective and nuanced feedback (related to feedback quality). This, in turn, enhances their self-efficacy in their abilities, encouraging further engagement in and skill development of providing feedback. This cycle boosts individual performance on feedback quality and self-efficacy in providing ChatGPT-assisted feedback. (No.29, p. 12) As for the operational domain, practical skills and strategies are crucial for effective technology use. When PSTs acquire knowledge about effectively employing technology like ChatGPT, they become more proficient, which enhances their confidence and improves practice quality (Herro et al., 2021). This highlights the reciprocal relationship between PSTs' feedback quality and self-efficacy, where improvements in cognitive understanding and operational skills mutually reinforce each other. (No. 29, p .13)
Synthesis 6 Institutional vision	AI-integrated curriculum (C10)	AI courses (SC26)	AI courses with clearly defined weekly or modular arrangements and a progressive learning pathway.	This workshop is structured to provide participants with a comprehensive understanding of generative AI. It addresses foundational principles, explores practical implications, and introduces prompting techniques to empower educators in utilizing AI effectively. The following sections outline the tool developed (CollectiveGPT at ed-tech.app) and topics covered, compactly visualized in Table 2. The workshop employs a scripted learning workshop design, which is intentionally structured to guide participants through progressive learning phases that lead to meaningful critical insights. These sudden realizations deepen understanding, often referred to as "eureka moments". We want participants to encounter unexpected scenarios that challenge their assumptions, create cognitive dissonance, and trigger

				deeper reflection. (No. 4, p. 9-10) The workshop design is intentionally structured to encourage participants to walk "in the agent's shoes" (Druga et al., 2019, as described in Design Consideration 2 in Long & Magerko, 2020), thereby fostering an immersive experience in which they perceive themselves as part of the collective human intelligence in decision-making and reasoning processes. (No. 4, p. 17) Session 1 introduced core concepts and ethical guidelines of generative AI (LO1–LO2). Session 2 practiced prompt-template families and iterative refinement techniques (LO3–LO4). Session 3 applied critical evaluation criteria to AI outputs and incorporated them into micro-lesson plans (LO5–LO6). By the end of the clinic, participants were expected to: (LO1) explain basic LLM architecture and limitations, (LO2) outline three ethical principles for classroom use, (LO3) compose prompts with role-task-context framing, (LO4) iteratively refine prompts to improve relevance and bias mitigation, (LO5) evaluate AI outputs against curricular standards, and (LO6) draft a lesson snippet that embeds AI-generated materials responsibly. A concise syllabus with activities and rubrics is provided in Appendix A, Table A1 to facilitate replication. (No. 6, p. 3)
		Educational frameworks (SC27)	Systematic design of AI teaching courses based on established frameworks (e.g., IBL, TPACK).	PTs' AI literacy development was supported through an AI-focused course designed following UNESCO's AI competency framework for teachers (Miao & Çukurova, 2024). (No. 15, p. 8) This course was specifically designed to equip them with the necessary skills to integrate AI tools, including ChatGPT, other AI technologies, into their instructional practices. In the first eight weeks, they were introduced to English teaching theories, methods, and AI-driven tools, and the throughout the rest of the course they were engaged in an IBL-based intervention focused on AI instructional integration in which they designed lesson plans, conducted microteaching, and demonstrated their AI-powered lessons for group observations and reflections. (No. 27, p. 4)

Commented [GU1]: More context needs to be brought into make the context of a course clear.

				To ensure fidelity, two doctoral students and two professors with expertise in writing research collaboratively developed a detailed workshop plan, outlining instructional objectives, procedures, and materials. The design of the workshop is grounded in Harris et al.'s (2009) TPACK framework. (No. 29, p. 5)
		Discipline-specific AI integration (SC28)	Design of differentiated AI training programs according to subject characteristics.	It would be essential that we also develop extended PD programs to support subject area teachers' AI literacy development and the integration of AI into subject area curricula. Yau et al. [26], with the focus of integrating AI education in K-12 education, conceptualized through their PD experience that the initial teacher PD could focus on AI knowledge and pedagogy enhancement whereas the advanced PD concerns integrating AI education into different subject areas. (No. 8, p. 11) Prompt engineering training should be part of teacher-education programs, e.g., as a last or additional part after each seminar or course, and it should be trained specifically with regard to subject-matter requirements. (No. 16, p. 28) Student teachers formed groups based on their specialization to develop lesson plans. Student teachers presented their lesson plans and received instant feedback from faculty and students. (No. 24, p. 23947)
	AI-focused training & incentives (C11)	AI-related professional learning (SC29)	Provision of ongoing professional learning opportunities to enable teachers to keep pace with the rapid development of AI.	Additionally, they can collaborate in faculty or faculty/student learning communities. Faculty and students learning together in their own courses can provide opportunities for students to be active participants in the process of learning about GenAI (Eaton, 2024). This is also supported by the Student AI Literacy (SAIL) framework by MacDowell et al. (2024), which prioritizes thinking and instructor support to learn about AI. (No. 23, p. 8) All of the participants indicated they need professional development and support to increase their GenAI competence. Robin said, 'Things have been changing very quickly. I have to update myself because this is new to me. And when I know more about this, I can help my student teachers use these tools or these, you know, applications more effectively pedagogically more effectively.' (No. 20, p. 7)

		Incentives (SC30)	Institutional incentive policies (e.g., certification, rewards, or career development opportunities) encouraging teachers to participate in AI training and instructional applications.	They can offer incentives such as paid GenAI learning sessions, subscriptions, micro-credentials or some certification that can be added to the teacher's CV. Administrators can support collaboration among institutions and educators to establish clear guidelines on the usage of AI and academic integrity. (No. 23, p.8)
	Institutional support (C12)	Institutional AI guidelines (SC31)	Development of institutional policies and ethical frameworks for AI use, accompanied by normative guidance established through cross-departmental collaboration.	Findings from this study show that adoption of GenAI tools is somewhat hindered by a lack of institutional and administrative guidelines, which are necessary for the diffusion process. The university wide policy seems to promote autonomy by emphasizing that each class should implement its own AI policy and communicate clear expectations but places the responsibility on faculty members, who may lack the essential knowledge about AI integration. This in turn, creates uncertainty and becomes a barrier to innovation (Rogers, 2003). (No. 23, p.7) Several of her PSTs reported using tools like ChatGPT for specific tasks, such as rephrasing lesson plan objectives or brainstorming activities. However, they expressed reluctance to rely on such tools for writing assignments due to concerns about potential reprimands. They desired clearer guidelines on how GenAI tools like ChatGPT could be appropriately integrated into their academic work. (No. 25, p. 10) Therefore, researchers and school teachers could collaborate to develop a guideline for using ChatGPT in writing feedback. This guide should include prompts tailored to writing contexts and scoring criteria, along with common issues encountered in ChatGPT interactions and corresponding solutions. (No. 29, p. 14)
		Leadership support (SC32)	Strategic support from institutional leadership, including resource provision and decision-making commitment, to promote the	The findings of this study reveal significant variability in AI preparedness and perceptions among K-12 educators. Educational leaders reported higher levels of readiness, reflecting their strategic roles and responsibilities in institutional planning. This aligns with previous studies indicating that leadership involvement is a critical factor in fostering innovation and technological integration in schools (Cheah et al.,

			systematic implementation of AI integration.	2025; Gestson & Core Team, 2024). Leaders were more likely to identify their institutions as being in advanced stages of AI planning or implementation, underscoring their role in driving organizational change. (No. 13, p. 13) There is a need to address institutional shortcomings, which can be done by establishing clear guidelines, providing dedicated resources, and implementing professional learning opportunities for faculty and students to facilitate the diffusion process. Without institutional and leadership support, the burden of AI integration falls on the individual educator, which can potentially lead to inconsistent adoption and missed opportunities to enhance teacher education. (No. 23, p. 7)